ARBITRARY COLORINGS OF EUCLIDEAN SPACE ARE NO EASIER THAN MEASURABLE ONES

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Abstract. A recent preprint released by OpenAI proves that every proper k-coloring of the plane, with arbitrary color classes, can be converted into a measurable k-coloring whose monochromatic unit pairs form a null set. It uses this to show that the plane is not 5-colorable. Its averaging step relies on the rotation group of the plane being abelian, and experts have noted that the corresponding step fails in dimension 3 and higher because the rotation group is not amenable.

We extend the transfer to every dimension $n \geq 2$. We average only over independent rotations of orthogonal coordinate 2-planes. We apply the planar rigidity theorem block by block through a polarization identity, and in odd dimensions we remove the remaining axis with an invariant-mean line average. Consequently

$$\chi(\mathbb{R}^n) \;\geq\; \lceil 1/m_1(\mathbb{R}^n) \rceil,$$

where $m_1(\mathbb{R}^n)$ is the maximal density of a measurable set avoiding distance 1. With published bounds on m_1 this gives $\chi(\mathbb{R}^3) \geq 7$ (previously 6), $\chi(\mathbb{R}^4) \geq 11$ (previously 9), $\chi(\mathbb{R}^8) \geq 53$ (previously 19), and $\chi(\mathbb{R}^n) \geq (1.268 + o(1))^n$, improving Raigorodskii's bound $(1.239 + o(1))^n$ from 2000.

The three-dimensional transfer and the deduction of $\chi(\mathbb{R}^3) \geq 7$ from the published density bound are formally verified in Lean 4. Applying the same argument to random independent sets, we show that finite unit-distance graphs attain the measurable density in every dimension, $\inf_G \alpha(G)/|G| = m_1(\mathbb{R}^n)$. In the plane this proves the first part of a recent conjecture of Dúcz and Varga: Erdős's function satisfies $f(N) = (m_1(\mathbb{R}^2) + o(1))N$.

1. Introduction

A proper k-coloring of $\mathbb{R}^n$ is a map $c\colon \mathbb{R}^n \to \{1, \dots, k\}$ with $c(x) \neq c(y)$ whenever $|x - y| = 1$. The chromatic number $\chi(\mathbb{R}^n)$ is the least such k, with no regularity imposed on the color classes. For the plane, Nelson and Isbell showed $4 \leq \chi(\mathbb{R}^2) \leq 7$ around 1950. De Grey raised the lower bound to 5 in 2018 [6]. A preprint released by OpenAI in September 2026 [11] proves $\chi(\mathbb{R}^2) \geq 6$, with a Lean formalization that we have rebuilt and checked. In three dimensions the best previous bounds were $6 \leq \chi(\mathbb{R}^3) \leq 15$ [10, 4, 14]. Asymptotically, $(1.239 + o(1))^n \leq \chi(\mathbb{R}^n) \leq (3 + o(1))^n$ [13, 8].

The measurable chromatic number $\chi_m(\mathbb{R}^n)$ requires measurable color classes. It is bounded below by $1/m_1(\mathbb{R}^n)$. Here

$$m_1(\mathbb{R}^n) = \sup\{\overline{\delta}(A) : A \subseteq \mathbb{R}^n \text{ measurable, no two points of } A \text{ at distance } 1\},$$

and $\overline{\delta}$ denotes upper density with respect to centered cubes. Fourier-analytic and semidefinite methods [12, 2, 5] give much stronger lower bounds for χ_m than combinatorial methods give for χ. For example, $\chi_m(\mathbb{R}^3) \geq 7$, and $\chi_m(\mathbb{R}^n) \geq (1.268 + o(1))^n$.

The OpenAI argument has two parts. The first is a transfer theorem [11, Theorem 1.3]: the plane has a proper k-coloring if and only if it has a *weak measurable* k-coloring. The second shows that weak measurable 5-colorings do not exist. The transfer averages a coloring of the algebraic points over algebraic translations and rotations. This needs the group $\mathbb{R}^2_{\mathrm{alg}} \rtimes SO(2)_{\mathrm{alg}}$

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to be amenable, which holds because it is solvable. Matolcsi, Ruzsa, Varga and Zsámboki [9] use the same kind of averaging for fractional colorings. They point out that it "cannot automatically be extended to higher dimensions," since the Følner property fails for Euclidean motions when $d \geq 3$.

Definition 1.1. A *weak measurable k-coloring* of $\mathbb{R}^n$ is a Lebesgue measurable $c\colon \mathbb{R}^n \to \{1, \dots, k\}$ with classes $A_i = c^{-1}(i)$ such that for every $R > 0$

$$\sum_{i=1}^{k} \int_{B(0,R)} \int_{S^{n-1}} \mathbf{1}_{A_i}(x)\, \mathbf{1}_{A_i}(x+u)\, d\sigma(u)\, dx = 0,$$

where σ is normalized surface measure on the unit sphere.

Theorem 1.2 (Transfer in every dimension)**.** *Let $n \geq 2$ and $k \geq 1$. If $\mathbb{R}^n$ has a proper k-coloring, then it has a weak measurable k-coloring.*

The converse holds by the density-point argument of [11, Lemma 4.3 and proof of Theorem 1.3], which is dimension-free. Thus $\chi(\mathbb{R}^n)$ equals the weak measurable chromatic number for every n.

Corollary 1.3. *If $\mathbb{R}^n$ has a proper k-coloring, then there are pairwise disjoint Borel sets $A_1^*, \dots, A_k^* \subseteq \mathbb{R}^n$, whose union has null complement, such that no A_i^* contains two points at distance* 1*. Consequently*

$$\chi(\mathbb{R}^n) \;\geq\; \lceil 1/m_1(\mathbb{R}^n) \rceil.$$

Corollary 1.4. $\chi(\mathbb{R}^3) \geq 7$*,* $\chi(\mathbb{R}^4) \geq 11$*,* $\chi(\mathbb{R}^5) \geq 17$*,* $\chi(\mathbb{R}^6) \geq 23$*,* $\chi(\mathbb{R}^7) \geq 39$*, and* $\chi(\mathbb{R}^8) \geq 53$*. Moreover*

$$\chi(\mathbb{R}^n) \;\geq\; (1.268 + o(1))^n \qquad (n \to \infty).$$

The bounds for $3 \leq n \leq 8$ use the density bounds of DeCorte, Oliveira and Vallentin [5]. These were published with exact verification programs, which we rebuilt and re-ran (Section 5). The asymptotic bound is the analytic Theorem 1.1 of Bachoc, Passuello and Thiery [2]. Table 1 lists further dimensions.

Theorem 1.5 (Formal verification)**.** *The case $n = 3$ of Theorem 1.2, in the form of Corollary 1.3, and the deduction "$m_1(\mathbb{R}^3) \leq 0.1532996 \Rightarrow \chi(\mathbb{R}^3) \geq 7$" are formally verified in Lean 4 with Mathlib. The proof depends only on the standard axioms* `propext`*,* `Classical.choice` *and* `Quot.sound`*.*

Section 7 describes the formal development. It builds on the Lean library accompanying [11], in particular on its formal proof of the planar rigidity theorem. The only input not formalized is the numerical density bound of [5], which appears as an explicit hypothesis.

Finally, the same method applied to random independent sets settles a question about finite graphs. For a finite unit-distance graph $G \subset \mathbb{R}^n$ let $\alpha_1(G) = \alpha(G)/|G|$, and set $\alpha_1(\mathbb{R}^n) = \inf_G \alpha_1(G)$. Averaging translates gives $m_1(\mathbb{R}^n) \leq \alpha_1(\mathbb{R}^n)$. In the plane, Ambrus et al. [1] and Matolcsi et al. [9] conjectured that the inequality is strict, "an interesting example of a natural infinite graph for which the measurable and the non-measurable variants of a parameter have different values." Dúcz and Varga [7] disproved the planar conjecture $\alpha_1(\mathbb{R}^2) = 1/4$, and conjectured instead that $f(N)/N = m_1(\mathbb{R}^2) + o(1)$, where $f(N)$ is the least independence number of an N-point unit-distance graph in the plane.

Theorem 1.6. *For every $n \geq 2$,*

$$\alpha_1(\mathbb{R}^n) = m_1(\mathbb{R}^n) \quad \textit{and} \quad \sup_G \chi_f(G) = \sup_G \chi_{gf}(G) = 1/m_1(\mathbb{R}^n),$$

where G ranges over finite unit-distance graphs in $\mathbb{R}^n$ and χ_{gf} is the geometric fractional chromatic number of [1, 9]. *In particular $f(N) = (m_1(\mathbb{R}^2) + o(1))N$, and $\sup_G \chi_f(G) > 4.0485$ in the plane.*

The second equality answers the question of [9] whether $\chi_{gf,0}(\mathbb{R}^d) = \chi_{f,0}(\mathbb{R}^d)$ for $d \geq 3$. As in [7], the graphs produced are not explicit in practice.

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2. Setting and inputs

Let $F = \overline{\mathbb{Q}} \cap \mathbb{R}$ be the field of real algebraic numbers, $E = F(i) \subset \mathbb{C}$, and $K = \{u \in E : |u| = 1\}$, a countable divisible abelian group. Write $n = 2m$ or $n = 2m + 1$ and identify $\mathbb{R}^n$ with $\mathbb{C}^m$ or $\mathbb{C}^m \times \mathbb{R}$. The algebraic points form the countable dense subgroup

$$V = E^m \quad \text{or} \quad V = E^m \times F.$$

The group K^m acts on V by rotating the j-th coordinate plane by the j-th component and fixing the last axis. These maps are linear isometries of $\mathbb{R}^n$. The group $G = V \rtimes K^m$ is solvable, hence amenable.

For a unitary representation U of a discrete group A that is a dense subgroup of a Euclidean space (here E, F or V), let

$$\mathcal{H}_c(U) = \{v : \ \|U_z v - v\| \to 0 \text{ as } z \to 0 \text{ in the Euclidean topology}\}.$$

This is a closed subspace. We use three results of [11].

Theorem 2.1 ([11, Theorem 2.3], Hilbert-space form)**.** *Let $U\colon E \to \mathcal{U}(\mathcal{H})$ and $W\colon K \to \mathcal{U}(\mathcal{H})$ be unitary representations with $W_k U_z = U_{kz} W_k$. If $v \perp \mathcal{H}_c(U)$ and $W_k v = v$ for all $k \in K$, then $\langle v, U_z v\rangle = 0$ for every $z \in E \setminus \{0\}$.*

In the Lean library of [11] this is `PlaneFiveColor.Spectral.residual_fourier`. It is deduced there from the rigidity of K-invariant probability measures on the dual of E that give no mass to the continuous characters.

Lemma 2.2 ([11, Lemma 4.1])**.** *If U is the Koopman representation of a measure-preserving action of A on a probability space, then $\mathcal{H}_c(U)$ is the L^2 space of a factor. Hence its orthogonal projection is a conditional expectation: it preserves $0 \leq f \leq 1$ and fixes constants. Translations act continuously on $\mathcal{H}_c(U)$ and extend to a strongly continuous, positivity-preserving unitary representation of the ambient Euclidean space.*

The proof in [11] is stated for $A = E$ but uses only that $\mathcal{H}_c$ is closed, contains constants, and is stable under products of bounded elements and under Lipschitz maps. It applies verbatim to $A = F$ and $A = V$.

Lemma 2.3 ([11, §2.2])**.** *There is a positive, F-translation-invariant linear functional I on bounded, boundedly supported functions $F \to \mathbb{C}$ with $I(\varphi|_F) = \int_{\mathbb{R}} \varphi$ for all $\varphi \in C_c(\mathbb{R})$. It integrates bounded Hilbert-valued functions weakly, with $\|I(\Phi)\| \leq I(\|\Phi\|)$.*

3. Two lemmas

Lemma 3.1. *Let U be a unitary representation as above and Y a bounded operator commuting with every U_z. Then $Y\mathcal{H}_c(U) \subseteq \mathcal{H}_c(U)$. If moreover Y is unitary, or self-adjoint, then $Y\mathcal{H}_c(U)^\perp \subseteq \mathcal{H}_c(U)^\perp$, and Y commutes with the orthogonal projection onto $\mathcal{H}_c(U)$.*

Proof. $\|U_zYv - Yv\| = \|Y(U_zv - v)\| \le \|Y\|\,\|U_zv - v\|$. In the unitary or self-adjoint case Y^* also commutes with U, so $\langle Yw, h\rangle = \langle w, Y^*h\rangle = 0$ for $w \perp \mathcal{H}_c$ and $h \in \mathcal{H}_c$. Splitting $v = Pv + (v - Pv)$ gives $PYv = YPv$. □

Lemma 3.2 (Twisted rigidity)**.** *Let U, W be as in Theorem 2.1, and let Y be unitary and commute with every U_z and every W_k. If $v \perp \mathcal{H}_c(U)$ is W-fixed, then $\langle v, U_zYv\rangle = 0$ for every $z \in E \setminus \{0\}$.*

Proof. By Lemma 3.1, Yv is W-fixed and orthogonal to $\mathcal{H}_c(U)$, and so is $v + sYv$ for every $s \in \mathbb{C}$. Theorem 2.1 applied to v, Yv and $v + sYv$ gives

$$0 = s\,\langle v, U_zYv\rangle + \bar{s}\,\langle Yv, U_zv\rangle \qquad (s \in \mathbb{C}).$$

Taking $s = 1$ and $s = i$ shows that both coefficients vanish. □

4. Proof of Theorem 1.2

We prove a slightly more general statement, which is also used in Section 6.

Proposition 4.1 (One-label transfer)**.** *Let μ be a G-invariant Borel probability measure on a compact metrizable G-space Ω. Let T_a $(a \in V)$ and R_κ $(\kappa \in K^m)$ be the Koopman operators on $\mathcal{H} = L^2(\Omega, \mu)$, so that $R_\kappa T_a R_\kappa^{-1} = T_{\kappa a}$. Let $f \in \mathcal{H}$ satisfy $0 \le f \le 1$, $R_\kappa f = f$ for all κ, and*

$$\langle f, T_a f\rangle = 0 \qquad \text{for every } a \in V \text{ with } |a| = 1.$$

Let Q be the orthogonal projection onto $\mathcal{H}_c(T)$, the vectors continuous for the whole V-action. Then $p = Qf$ satisfies $0 \le p \le 1$, $\mathbb{E}p = \mathbb{E}f$, and $\langle p, T_x p\rangle = 0$ for every $x \in \mathbb{R}^n$ with $|x| = 1$, where T_x is the extension of Lemma 2.2.

Proof. For a block $j \le m$ let $U_z^{(j)} = T_{\iota_j z}$, where $\iota_j z$ places $z \in E$ in the j-th coordinate plane. Let $W^{(j)}$ be the rotations of that plane, and P_j the projection onto $\mathcal{H}_c(U^{(j)})$. In odd dimension let $S_t = T_{(0,\dots,0,t)}$ for $t \in F$, and let P_0 be the projection onto $\mathcal{H}_c(S)$.

Commutation. All translations commute with one another and with rotations of the other blocks, and axis translations commute with all rotations. By Lemma 3.1, each P_j (and P_0) commutes with every T_a, with the rotations of the other blocks, and, through the self-adjoint case of Lemma 3.1, with the other projections. Covariance gives $\|U_z^{(j)}R_\kappa v - R_\kappa v\| = \|U^{(j)}_{\kappa^{-1}z}v - v\|$, so P_j also commutes with the rotations of its own block. A vector continuous in every block and along the axis is continuous for V, so

$$Q = P_m \cdots P_1 \quad \text{or} \quad Q = P_0P_m \cdots P_1.$$

Each factor is a conditional expectation (Lemma 2.2). Hence $0 \le p \le 1$, and $\mathbb{E}p = \langle f, Q1\rangle = \mathbb{E}f$.

The plane blocks. Call a unit vector $a \in V$ *generic* if all of its block components $a_j \in E$ are nonzero. Let $g^{(0)} = f$ and $g^{(j)} = P_j g^{(j-1)}$; these are rotation-fixed. Put $v = (1 - P_j)g^{(j-1)}$, which is $W^{(j)}$-fixed and orthogonal to $\mathcal{H}_c(U^{(j)})$. Write $T_a = U^{(j)}_{a_j}Y$, where $Y = T_{a - \iota_j a_j}$ commutes with $U^{(j)}$ and $W^{(j)}$. Lemma 3.2 gives $\langle v, T_a v\rangle = 0$. The cross terms vanish by Lemma 3.1, so

$$\langle g^{(j)}, T_a g^{(j)}\rangle = \langle g^{(j-1)}, T_a g^{(j-1)}\rangle \qquad (a \text{ generic}).$$

With $g = g^{(m)}$ this gives $\langle g, T_a g\rangle = \langle f, T_a f\rangle = 0$ for generic a.

Even n. Here $g = p$. The rational points are dense in S^{n-1} (inverse stereographic projection), and the non-generic ones lie in finitely many proper subspheres. So generic algebraic unit vectors are dense, and continuity of $x \mapsto \langle p, T_x p\rangle$ finishes the proof.

Odd n: the axis. Let $w = g - p \perp \mathcal{H}_c(S)$. Both p and w lie in the range of $P_m \cdots P_1$, so block translations act continuously on them and extend to real arguments. Since P_0 commutes with T_a,

$$\langle p, T_a p\rangle = -\langle w, T_a w\rangle \qquad (a \text{ generic}). \tag{1}$$

Fix an algebraic direction u in the plane blocks with all block components nonzero, and let $\rho(t) = \sqrt{1-t^2}$. For $t \in F \cap (-1,1)$ the vector $a(t) = (\rho(t)u, t)$ is a generic algebraic unit vector, because F is real closed. Let $b(t) = \langle p, T_{(\rho(t)u,t)}p\rangle$; it is continuous in real t.

Fix a real $\varphi \in C_c((-1,1))$. The window $\Phi(t) = \varphi(t)\, T_{(\rho(t)u,0)}w$ is continuous with compact support, hence uniformly continuous. Let

$$\mathcal{V} = I_t\big(S_t\Phi(t)\big).$$

Every $S_t\Phi(t)$ lies in the closed subspace $\mathcal{H}_c(S)^\perp$, so $\mathcal{V}$ does too. By translation invariance of I,

$$\|S_s\mathcal{V} - \mathcal{V}\| \le I_t\big(\|\Phi(t-s) - \Phi(t)\|\big) \to 0 \quad (s \to 0),$$

so $\mathcal{V} \in \mathcal{H}_c(S) \cap \mathcal{H}_c(S)^\perp = \{0\}$. Pairing $\mathcal{V}$ with $c\,w$ ($c \in \mathbb{C}$) and using (1) and Lemma 2.3,

$$0 = I_t\Big(\varphi(t)\,\mathrm{Re}\big(\bar c\,\langle w, T_{a(t)}w\rangle\big)\Big) = -\int_{\mathbb{R}} \varphi(t)\,\mathrm{Re}\big(\bar c\, b(t)\big)\,dt.$$

Taking $c \in \{1, i\}$ and applying the fundamental lemma of the calculus of variations gives $b \equiv 0$ on $(-1,1)$. Algebraic directions are dense, so by continuity $\langle p, T_x p\rangle = 0$ for every real unit vector x. □

Proof of Theorem 1.2. Restrict the given coloring to V. Let $\Omega \subseteq \{1,\dots,k\}^V$ be the nonempty, closed, G-invariant set of proper labelings of the unit-distance graph on V, and let μ be a G-invariant probability on Ω, obtained by Følner averaging. Let $f_i = \mathbf{1}[\ell(0) = i]$. These are rotation-fixed, satisfy $\sum_i f_i = 1$, and $\langle f_i, T_a f_i\rangle = \mu\{\ell(0) = i = \ell(a)\} = 0$ for algebraic unit a. Proposition 4.1 gives $p_i = Qf_i \in [0,1]$ with $\sum_i p_i = Q1 = 1$ and zero correlation at every real unit vector.

Choose jointly measurable fields $H_i(x,\omega) \in [0,1]$ representing $T_x p_i$ for almost every x, as in [11, proof of Proposition 4.2]. Then

$$\int_\Omega \sum_i \int_{B(0,R)} \int_{S^{n-1}} H_i(x,\omega) H_i(x+u,\omega)\, d\sigma\, dx\, d\mu = |B(0,R)| \sum_i \int_{S^{n-1}} \langle p_i, T_u p_i\rangle\, d\sigma = 0.$$

Fix ω such that all these inner integrals vanish for rational R, and such that $\sum_i H_i(\cdot,\omega) = 1$ almost everywhere. Color each x by the least i with $H_i(x,\omega) > 0$. This coloring is measurable, and its monochromatic unit pairs form a $dx\,d\sigma$-null set. □

Proof of Corollary 1.3. Let A_i^* be the set of Lebesgue density points of $A_i = c^{-1}(i)$. These sets are Borel and pairwise disjoint, and their union is conull by the Lebesgue differentiation theorem. By [11, Lemma 4.3], whose proof is dimension-free, A_i^* contains no two points at distance 1. Since $\sum_i |A_i^* \cap [-T,T]^n| = (2T)^n$ for every T, some class has upper density at least $1/k$. Hence $1/k \le m_1(\mathbb{R}^n)$. □

5. Numerical inputs

Table 1 combines Corollary 1.3 with published bounds for $m_1(\mathbb{R}^n)$. The rows $3 \le n \le 8$ come from [5, Table 2]. DeCorte, Oliveira and Vallentin provide the dual solutions and an exact verifier (`rn_verify`, GMP/MPFR) as ancillary files. We rebuilt the verifier, after one portability fix that makes a base-class destructor virtual and does not affect the arithmetic, and re-ran it on the published data. For $n = 3$ it reports "Solution checked; bound = 0.153299568560593". The patched verifier and logs are in the accompanying repository,

https://github.com/outsidetext/euclidean-chromatic-transfer. The rows $n \geq 9$ come from [2, Table 3]. They rest on floating-point linear programs that we have not independently verified; the margin at $n = 24$ is about 0.02. The asymptotic row is the analytic Theorem 1.1 of [2]. Previous lower bounds for $\chi(\mathbb{R}^n)$ are as collected in [3].

n	$m_1(\mathbb{R}^n) \leq$	previous $\chi(\mathbb{R}^n) \geq$	new $\chi(\mathbb{R}^n) \geq$
3	0.1532996	6	7
4	0.0985701	9	11
5	0.0624485	9	17
6	0.0450325	11	23
7	0.0260782	15	39
8	0.0190945	19	53
9–12	[2]	22, 30, 35, 37	69, 90, 116, 164
24	$1.84366 \cdot 10^{-4}$	–	5424
$n \to \infty$	$(1.268 + o(1))^{-n}$	$(1.239 + o(1))^n$	$(1.268 + o(1))^n$

TABLE 1. Lower bounds for $\chi(\mathbb{R}^n)$ from Corollary 1.3.

6. Finite graphs attain the measurable density

Proof of Theorem 1.6. (i) $m_1 \leq \alpha_1$. Let A be measurable and unit-distance-free, with densities $d_j \to d$ along centered cubes Q_j. For a finite unit-distance graph G,

$$\frac{1}{|Q_j|}\int_{Q_j} |(A-x)\cap G|\,dx = \sum_{g\in G}\frac{|A\cap(Q_j+g)|}{|Q_j|} \to d|G|,$$

so some translate of A meets G in an independent set of size at least $(d-\varepsilon)|G|$. (This also shows that the definition of m_1 via $\inf_X \sup_A \lambda(A)/\lambda(X)$ used in [9] agrees with ours.)

(ii) $\sup_G \chi_f(G) \geq 1/m_1$. Suppose every finite subgraph of the unit-distance graph on V has $\chi_f \leq w$. Normalizing optimal fractional colorings gives, for each finite $H \subset V$, a random independent set of H covering each vertex with probability at least $1/w$. A weak-$*$ limit in $\{0,1\}^V$, in which cylinder events are clopen, yields a random independent set I of V with $\mathbb{P}(v \in I) \geq 1/w$ for all v. Følner averaging over G makes its law invariant without affecting these properties.

Apply Proposition 4.1 to $f = \mathbf{1}[0 \in I]$, and let $H(x,\omega)$ be jointly measurable versions of $T_x p$. For almost every ω, $H(x)H(x+u) = 0$ for $dx\,d\sigma$-almost every (x,u). The cube averages $X_T(\omega) = |Q_T|^{-1}\int_{Q_T} H(x,\omega)\,dx \in [0,1]$ have mean $\mathbb{E}p \geq 1/w$, so by reverse Fatou $\mathbb{P}(\limsup_T X_T \geq 1/w) > 0$. For such an ω, the density points of $\{H(\cdot,\omega) > 0\}$ form a measurable unit-distance-free set of upper density at least $1/w$. Hence $w \geq 1/m_1$.

(iii) The proof of [9, Theorem 2], $\sup_G \chi_f(G) = 1/\alpha_1$, uses only translations through a lattice cube and holds verbatim in $\mathbb{R}^n$. Together with (i) and (ii) this gives $\alpha_1 = m_1$.

Averaging a measurable unit-distance-free set over random translates in large cubes and a Haar-random element of $O(n)$ shows $\chi_{gf}(G) \leq 1/m_1$ for every finite G. Since $\chi_f \leq \chi_{gf}$, all three suprema coincide.

Finally, f is subadditive (place copies far apart), so Fekete's lemma gives $f(N)/N \to \inf_N f(N)/N = \alpha_1(\mathbb{R}^2) = m_1(\mathbb{R}^2)$. The value 4.0485 uses $m_1(\mathbb{R}^2) \leq 0.24699$ [1]. □

7. Formal verification

The Lean development, available at https://github.com/outsidetext/euclidean-chromatic-transfer (about 1,800 lines, Lean 4.34.1, Mathlib commit `d13f23b`) builds on the 70 modules of the library of [11] on which its planar result depends. It models $\mathbb{R}^3$ as $\mathbb{C}\times\mathbb{R}$ with $|(z,t)|^2=|z|^2+t^2$. The main statements are:

```
theorem three_space_transfer {k : ℕ} [NeZero k] (c : ℂ × ℝ → Fin k)
    (hc : ∀ x y : ℂ × ℝ, norm (x.1 - y.1)^2 + (x.2 - y.2)^2 = 1 → c x ≠ c y) :
    ∃ A : Fin k → Set (ℂ × ℝ), (∀ i, MeasurableSet (A i)) ∧
      Pairwise (fun i j => Disjoint (A i) (A j)) ∧
      volume (Set.iUnion A)ᶜ = 0 ∧
      ∀ i, ∀ x ∈ A i, ∀ y ∈ A i, norm (x.1 - y.1)^2 + (x.2 - y.2)^2 ≠ 1

theorem no_six_coloring_of_density (hDOV : DensityBound 0.1532996) :
    ¬ ∃ c : ℂ × ℝ → Fin 6, ∀ x y, usq (x - y) = 1 → c x ≠ c y
```

Here `DensityBound` δ states that every measurable unit-distance-free subset of $\mathbb{C}\times\mathbb{R}$ has upper density at most δ along the cubes $[-T,T]^3$. The modules follow Sections 2–4:

- `Model`: the label space over $E\times F$ and its invariant law.
- `Rigidity`: Lemmas 3.1 and 3.2, and the plane step.
- `Smoothing` and `LineMain`: the vector-window invariant integral and the axis step.
- `Sampling` and `Density`: measurable fields, the sampled coloring, density points.
- `Corollary`: the deduction from `DensityBound`.

`#print axioms` reports only `propext`, `Classical.choice` and `Quot.sound`.

8. Remarks

Why the torus suffices. Theorem 1.2 needs neither invariance under all of $SO(n)$ nor a disintegration of spectral measures. A maximal torus K^m of algebraic rotations is abelian, so averaging over it is possible. Rigidity applied to one block at a time, via Lemma 3.2, then removes every spectral component that is discontinuous in some block. The remaining axis in odd dimension is handled by the invariant mean, as in [11, Lemma 2.5].

Other forbidden distances. The argument uses only that the forbidden distance is algebraic. It therefore transfers proper colorings avoiding any finite set of algebraic distances, and gives $\chi\geq 1/m_D$ for the corresponding densities.

The plane. By [11, Theorem 1.3], deciding between $\chi(\mathbb{R}^2)=6$ and 7 is equivalent to deciding whether weak measurable 6-colorings of the plane exist. Density and fractional methods cannot settle this, since by Theorem 1.6 they are capped at $1/m_1(\mathbb{R}^2)\leq 1/0.2293<4.37$.

References

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